

KLOTEN, Sweden

WMO: 024310

Lat: 59.8703N

Lon: 15.2522E

Elev: 289

StdP: 97.90

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-15.0	-21.2	0.6	-17.0	-17.4	0.8	-14.4	7.3	0.1	6.4	-0.1	0.8	30	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	25.4	16.9	23.6	16.1	21.8	15.3	18.5	22.8	17.4	21.4	16.5	20.1	2.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.6	19.2	16.1	11.8	18.4	15.0	11.0	18.0	53.4	23.1	50.2	21.5	47.3	19.9	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.4	4.9	DB	-22.7	27.6	4.5	1.7	-25.9	28.8	-28.5	29.8	-31.1	30.8	-34.3	32.1	(4)
(5)			WB	-22.4	19.6	4.5	1.6	-25.7	20.7	-28.3	21.7	-30.8	22.5	-34.0	23.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-5.3	-4.2	-0.8	3.9	9.7	13.3	16.2	14.4	10.6	4.7	0.6	-2.8	(6)	
	DBStd	8.28	4.85	4.99	4.46	3.66	4.29	2.95	3.01	2.50	2.69	3.22	3.74	5.10	(7)	
	HDD10.0	2327	473	397	336	186	59	5	0	1	24	166	281	398	(8)	
	HDD18.3	4861	732	631	594	433	268	153	83	126	232	422	531	657	(9)	
	CDD10.0	529	0	0	0	2	50	105	191	136	42	3	0	0	(10)	
	CDD18.3	22	0	0	0	0	0	3	16	3	0	0	0	0	(11)	
	CDH23.3	182	0	0	0	0	10	27	125	21	0	0	0	0	(12)	
	CDH26.7	19	0	0	0	0	0	1	16	3	0	0	0	0	(13)	
(14) Wind	WSAvg	2.3	2.3	2.4	2.5	2.4	2.4	2.4	2.2	2.1	2.3	2.4	2.4	2.4	(14)	
(15) Precipitation	PrecAvg	730	55	41	33	44	58	70	85	83	61	70	66	60	(15)	
	PrecMax	1029	107	92	75	126	136	147	190	187	124	182	170	110	(16)	
	PrecMin	531	6	9	0	0	18	3	12	22	9	0	20	14	(17)	
	PrecStd	96	26	20	20	27	29	35	32	35	37	44	34	30	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	6.9	13.4	18.8	24.8	26.3	28.5	25.9	20.6	15.0	9.5	8.1	(19)	
		MCWB	4.6	3.2	7.9	9.9	14.3	16.2	17.9	18.0	14.9	11.8	8.9	7.7	(20)	
	2%	DB	3.5	4.3	9.2	16.3	22.5	23.9	26.5	22.9	18.2	12.6	7.8	6.7	(21)	
		MCWB	3.1	2.8	5.2	8.8	13.5	15.8	17.9	16.9	14.2	11.1	7.5	6.2	(22)	
	5%	DB	2.2	3.0	7.2	13.2	20.4	21.7	24.6	20.9	16.5	11.1	6.7	4.9	(23)	
		MCWB	1.9	2.0	3.5	7.7	12.9	14.8	17.0	15.6	13.4	10.1	6.3	4.3	(24)	
	10%	DB	0.9	1.9	5.2	10.6	17.9	19.5	22.6	19.3	15.0	9.8	5.8	3.3	(25)	
		MCWB	0.6	1.2	2.4	5.9	12.1	13.4	16.2	14.9	12.4	8.6	5.5	2.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	4.3	7.8	10.7	16.0	18.0	20.2	19.2	16.4	12.8	8.9	7.9	(27)	
		MCDB	4.8	5.0	13.0	17.6	21.3	22.4	24.3	22.9	18.5	13.9	9.0	8.1	(28)	
	2%	WB	3.2	3.1	5.6	9.3	14.9	16.8	19.1	17.9	15.1	11.7	7.5	6.3	(29)	
		MCDB	3.4	3.9	8.7	15.0	20.0	21.3	23.7	21.1	17.5	12.3	7.8	6.7	(30)	
	5%	WB	2.0	2.2	4.2	8.0	13.8	15.7	18.0	16.8	14.0	10.3	6.5	4.4	(31)	
		MCDB	2.3	2.9	6.3	12.3	18.6	20.3	22.6	19.4	15.5	10.9	6.7	4.7	(32)	
	10%	WB	0.8	1.2	2.9	6.7	12.5	14.4	17.1	15.9	13.0	8.8	5.6	2.8	(33)	
		MCDB	1.0	1.8	4.6	9.7	16.9	18.6	21.4	18.2	14.4	9.3	5.8	3.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	5.6	7.9	9.4	10.0	9.2	9.3	8.4	7.2	5.6	3.8	4.6	(35)	
		MCDBR	4.4	6.6	9.8	13.9	13.7	12.8	12.4	11.1	9.2	6.7	4.8	5.1	(36)	
	5% WB	MCWBR	4.3	5.0	6.2	7.5	6.8	6.1	5.0	5.6	5.4	5.1	4.5	5.5	(37)	
		MCWBR	4.5	5.1	9.0	12.2	11.8	10.9	10.8	9.2	8.1	6.1	4.8	5.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.255	0.298	0.321	0.357	0.355	0.347	0.368	0.361	0.329	0.317	0.279	0.234	(40)	
		taud	2.216	2.313	2.381	2.367	2.431	2.473	2.429	2.444	2.525	2.468	2.283	2.120	(41)	
	Ebn at Noon		541	713	816	844	873	886	855	833	807	692	504	406	(42)	
		Edh at Noon	40	65	83	101	101	99	101	93	73	56	38	29	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.29	0.92	2.45	3.87	4.86	5.53	5.02	3.93	2.61	1.17	0.37	0.19	(44)	
	RadStd		0.05	0.17	0.29	0.45	0.48	0.37	0.59	0.43	0.34	0.21	0.09	0.02	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (60 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-119	N/A	N/A	N/A	

Nomenclature: See separate page